

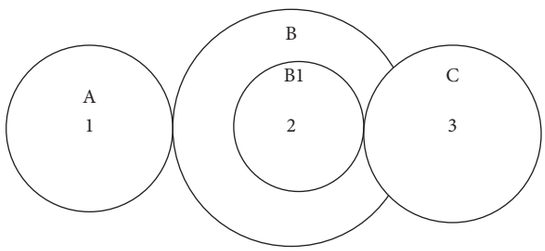
Compound Gear Train

Two gears mounted in the same shaft. So rotational speed are same. ($N_2 = N_3$)

Velocity ratio: $\frac{N_4}{N_1} = \frac{T_1}{T_2} \times \frac{T_3}{T_4}$

EXAMPLE 7

A compound gear train is arranged in which the driving shaft 1 and shaft 3 through intermediate shaft 2. The gear A on shaft 1 has 40 teeth which rotates at 900 rpm in clock wise direction. It meshes with a gear B of 80 teeth on shaft 2. This shaft has another gear with 50 teeth which meshes with a gear, C on shaft 3. what is the speed and direction of rotation of the gear on shaft 3 which has 60 teeth?



T_A = 40teeth T_B = 80teet T_C = 60teeth

N_A = 900rpm T_{B1} = 50teeth

Number of teeth on the driving gear (T_A) = 40 teeth

Number of teeth on the intermediate shaft gear (T_B) = 80 teeth

Number of teeth on the intermediate shaft gear (T_{B1}) = 50 teeth

Number of teeth on the driver gear (T_C) = 60 teeth

Number of rotational of driving gear (N_A) = 900 rpm

$$N_B = \frac{T_A N_A}{T_B}$$

$$N_B = \frac{40 \times 900}{80}$$

N_B = 450 RPM

Two gears (N_B and N_{B1}) mounted in the same shaft. So rotational speed are same.

$N_B = N_{B1}$ = 450 RPM

$$N_C = \frac{T_{B_1} \times N_{B_2}}{T_c}$$

$$= \frac{50 \times 450}{60}$$

= 375 RPM

N_C = 375 RPM Anti clock wise direction

Advantages of gears

1. By using gear train, large velocity ratio can be obtained with minimum space
2. Gears are mechanically strong, So higher loads can be lifted
3. They are used for positive drive, so its velocity ratio remains constant.
4. Gears require only lubrication, hence less maintenance is required.

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